SHUPENG NING

Tel: +1(571) 519-9918 ◇ Email: spning@utexas.edu ◇ Google Scholar ◇ Homepage

RESEARCH INTERESTS

Emerging Photonic-Electronic Platform for Efficient AI Computing

- Photonic and mixed-signal computing for AI acceleration
- Heterogeneous electronic-photonic integration and hardware-efficient computing architectures
- Cross-layer and domain-specific co-design for emerging computing systems

Integrated Photonic & Multiphysics Microsystems

- Optical biosensing and sample-to-answer systems for point-of-care bioanalysis
- Multiphysics microsystems for sensing, actuation, and information processing

EDUCATION AND TRAINING

California Institute of Technology, Pasadena, CA, USA *May 2026 – Present*
Postdoctoral Scholar in Medical Engineering and Electrical Engineering
Advisor: Prof. Lihong V. Wang

The University of Texas at Austin, Austin, TX, USA *Aug. 2021 – Apr. 2026*
Ph.D. in Electrical and Computer Engineering
Advisor: Prof. Ray T. Chen; Co-advisor: Prof. David Z. Pan
Cumulative GPA: 4.0/4.0

Nankai University, Tianjin, China *Sep. 2016 – Jun. 2020*
B.S. in Microelectronics Science and Engineering
Cumulative GPA: 92.63/100 (3.91/4.0); Major GPA: 93.9/100 (3.99/4.0)

AWARDS AND HONORS

2026 **DAC Young Fellow**, ACM/IEEE Design Automation Conference
2025 **Ben Streetman Prize**, The University of Texas at Austin
2025 **Dr. Jacob A. Abraham Graduate Fellowship**,
Department of Electrical and Computer Engineering, The University of Texas at Austin
2024 **Robert S. Hilbert Memorial Optical Design Competition Award**, Synopsys
2023 **Harry Philip Whitworth Endowed Graduate Fellowship**,
Department of Electrical and Computer Engineering, The University of Texas at Austin
2020 **Outstanding Graduate**, Nankai University
2019 **Zhou Enlai Scholarship**, Nankai University, highest honor awarded to ten undergraduates each year
2019 **First Prize Scholarship**, Nankai University, Top 5%
2018 **National Scholarship**, Ministry of Education of China, Top 2%
2018 **Honorable Mention in Mathematical Contest in Modeling**,
Consortium for Mathematics and Its Applications (COMAP)
2018 **Merit Student**, Nankai University, Top 1%
2017 **Tianjin Municipal Government Scholarship**, Tianjin Municipal Government, Top 2.5%

TEACHING & MENTORING

Teaching Assistant / Guest Lecturer — ECE 383P-8: Optical Communications — *Spring 2025*

- *Lecture topic: Integrated Photonics for Interconnects, Computing, and Sensing*

Graduate Student Mentor — EE 464: Senior Design Project — *Fall 2021, Fall 2022*

- *Mentored two undergraduate project teams on integrated photonic biosensing*

PROFESSIONAL EXPERIENCE

Applied Materials, CA, USA | CTO Team, Intern — *May 2025 – Aug. 2025*

- Modeled, designed, and characterized advanced devices and integration schemes for next-generation photonic interconnects
- Drove feasibility assessments and performance/yield improvements through cross-functional collaboration

Intel Labs, Beijing, China | Intern — *Dec. 2020 – Jul. 2021*

- Deployed spiking neural networks on the Intel *Loihi* neuromorphic processor for gesture/posture recognition and health-related sensing
- Implemented the full pipeline from neural encoding and training to on-chip mapping for real-time inference

SELECTED PUBLICATIONS

Journal Papers

[J12] **Shupeng Ning**, Chenghao Feng, Zhenxiang Xu, Hanqing Zhu, David Z Pan, Jiaqi Gu, and Ray T Chen. "General-Purpose Photonic Computing Primitive for Contemporary Artificial Intelligence". *arXiv:2605.3051 (under minor revision for Nature Communications)*, 2026.

[J11] Hanqing Zhu, **Shupeng Ning**, Hongjian Zhou, Ziang Yin, Ray T Chen, Jiaqi Gu, and David Z Pan. "Harnessing Photonics for Machine Intelligence". *IEEE Circuits and Systems Magazine* 26(3), pp. 46–74, Aug. 2026.

[J10] **Shupeng Ning**, Hanqing Zhu, Chenghao Feng, Jiaqi Gu, David Z Pan, and Ray T Chen. "Hardware-Efficient Photonic Tensor Core: Accelerating Deep Neural Networks with Structured Compression". *Optica* 12(7), pp. 1079–1089, Jul. 2025.

[J9] Sourabh Jain, May H Hlaing, Kang-Chieh Fan, Jason Midkiff, **Shupeng Ning**, Chenghao Feng, Po-Yu Hsiao, Patrick T Camp, and Ray T Chen. "Incubating Advances in Integrated Photonics with Emerging Sensing and Computational Capabilities". *Applied Physics Reviews* 12(1), Feb. 2025.

[J8] **Shupeng Ning**, Hanqing Zhu, Chenghao Feng, Jiaqi Gu, Zhixing Jiang, Zhoufeng Ying, Jason Midkiff, Sourabh Jain, May H Hlaing, David Z Pan, et al. "Photonic-Electronic Integrated Circuits for High-Performance Computing and AI Accelerators". *Journal of Lightwave Technology* 42(22), pp. 7834–7859, Jul. 2024.

[J7] Shanny Lin, **Shupeng Ning**, Hanqing Zhu, Tao Zhou, Christopher L Morris, Steven Clayton, Mathew J Cherukara, Ray T Chen, and Zhehui Wang. "Neural Network Methods for Radiation Detectors and Imaging". *Frontiers in Physics* 12, p. 1334298, Feb. 2024.

[J6] Chenghao Feng, Jiaqi Gu, Hanqing Zhu, **Shupeng Ning**, Rongxing Tang, May Hlaing, Jason Midkiff, Sourabh Jain, David Z Pan, and Ray T Chen. "Integrated Multi-Operand Optical Neurons for Scalable and Hardware-Efficient Deep Learning". *Nanophotonics* 13(12), pp. 2193–2206, Jan. 2024.

[J5] Kyoung Min Yoo, Kang-Chieh Fan, May Hlaing, Sourabh Jain, **Shupeng Ning**, Yue An, and Ray T Chen. "Lab-on-a-Chip Optical Biosensor Platform: A Micro-Ring Resonator Integrated with a Near-Infrared Fourier Transform Spectrometer". *Optics Letters* 48(20), pp. 5371–5374, Sep. 2023.

[J4] **Shupeng Ning**, Hao-Chen Chang, Kang-Chieh Fan, Po-Yu Hsiao, Chenghao Feng, Devan Shoemaker, and Ray T Chen. "A Point-of-Care Biosensor for Rapid Detection and Differentiation of COVID-19 Virus (SARS-CoV-2) and Influenza Virus Using Subwavelength Grating Micro-Ring Resonator". *Applied Physics Reviews* 10(2), May 2023.

[J3] **Shupeng Ning**, Shuchang Liu, Yunjie Xiao, Guanyu Zhang, Weiwei Cui, and Mark Reed. "A Microfluidic Chip with a Serpentine Channel Enabling High-Throughput Cell Separation Using Surface Acoustic Waves". *Lab on a Chip* 21(23), pp. 4608–4617, Nov. 2021.

[J2] Guanyu Zhang, Weiwei Cui, Wei Pang, Shuchang Liu, **Shupeng Ning**, Xingchen Li, and Mark Reed. "Programmable and Parallel Trapping of Submicron/Nanoscale Particles Using Acoustic Micro-Pillar Array Chip". *Advanced Materials Interfaces* 8(24), p. 2101334, Nov. 2021.

[J1] Wenjia Zhang, Qian Du, Zheng Dou, **Shupeng Ning**, Qingzhao Wang, Yuemeng Dong, Zhao Yue, Weixiang Ye, and Guohua Liu. "Ultrasensitive Detection of Lead (II) Ion by Dark-Field Spectroscopy and Glutathione-Modified Gold Nanoparticles". *Sensors and Actuators B: Chemical* 321, p. 128548, Oct. 2020.

Conference Papers

[C18] Shanny Lin, **Shupeng Ning**, CL Morris, Gary Saavedra, Zhaowen Tang, Zhehui Wang, and Ray T Chen. "Towards Machine-learning Integrated Optoelectronic Computing Chips for Proton Radiography". *CLEO: Applications and Technology*, JW1–171, May 2026.

[C17] **Shupeng Ning**, Hanqing Zhu, Chenghao Feng, Zhenxiang Xu, Jiaqi Gu, David Z Pan, and Ray T Chen. "Dynamic High-Speed Integrated Photonic Tensor Core for Full-Range Transformer Self-Attention". *CLEO: Applications and Technology*, ATU1E–4, May 2026.

[C16] Zhenxiang Xu, **Shupeng Ning**, Chenghao Feng, Hanqing Zhu, David Z Pan, Jiaqi Gu, and Ray T Chen. "Hardware-Efficient Photonic Tensor Cores Based on Differential Multi-Operand Mach-Zehnder Interferometers". *CLEO: Applications and Technology*, ATU1E–2, May 2026.

[C15] **Shupeng Ning**, Hanqing Zhu, Chenghao Feng, Zhenxiang Xu, David Z Pan, and Ray T Chen. "A Dynamically Operated Photonic Tensor Core for Attention Mechanisms in Transformer". *Optical Interconnects and Packaging 2026*, PC139030D, Mar. 2026.

[C14] Hanqing Zhu, Zhican Zhou, **Shupeng Ning**, Xuhao Wu, Ray T Chen, Yating Wan, and David Z Pan. "ENLighten: Lighten the Transformer, Enable Efficient Optical Acceleration". *2026 31st Asia and South Pacific Design Automation Conference (ASP-DAC)*, pp. 405–412, Jan. 2026.

[C13] **Shupeng Ning**, Hanqing Zhu, Ziang Yin, Chenghao Feng, Spencer Denton, David Z Pan, Jiaqi Gu, and Ray T Chen. "Microring-Based Multi-Operand Optical Neurons with On-Chip Trainable Nonlinearity". *CLEO: Applications and Technology*, AA120–1, May 2025.

[C12] Chun-Ju Yang, Hanqing Zhu, **Shupeng Ning**, Jiaqi Gu, Chenghao Feng, David Z Pan, and Ray T Chen. "Early Pancreatic Cancer Detection Using Deep Learning and Optical Neural Networks on Integrated Photonic Chips". *CLEO: Applications and Technology*, AA110–5, May 2025.

[C11] **Shupeng Ning**, Chenghao Feng, Jiaqi Gu, Hanqing Zhu, Rongxing Tang, David Z Pan, and Ray T Chen. "A Compact Optical Neuron Based on Multi-Operand Microring Resonators". *Optical Interconnects and Packaging 2025*, PC1337202, Mar. 2025.

[C10] Hanqing Zhu, Wenyan Cong, Guojin Chen, **Shupeng Ning**, Ray T Chen, Jiaqi Gu, and David Z Pan. "PACE: Pacing Operator Learning for Accurate Optical Field Simulation of Complicated Photonic Device". *Proceedings of the 38th International Conference on Neural Information Processing Systems*, pp. 67535–67555, Dec. 2024.

[C9] **Shupeng Ning**, Hanqing Zhu, Chenghao Feng, Christian Uselton, Jiaqi Gu, Rongxing Tang, David Z Pan, and Ray T Chen. "Realization of a Compact Photoelectric Platform for Optical Convolution Processing". *CLEO: Science and Innovations*, SM4M–2, May 2024.

[C8] Chun-Ju Yang, Hanqing Zhu, **Shupeng Ning**, Chenghao Feng, Jiaqi Gu, David Z Pan, and Ray T Chen. "Deep Learning Enhanced Early Detection of Pancreatic Cancer Using Integrated Photonic Chip Based Optical Neural Networks". *2024 Conference on Lasers and Electro-Optics (CLEO)*, pp. 1–2, May 2024.

[C7] **Shupeng Ning**, Jiaqi Gu, Chenghao Feng, Rongxing Tang, Hanqing Zhu, David Z Pan, and Ray T Chen. "A Hardware-Efficient Silicon Electronic-Photonic Chip for Optical Structured Neural Networks". *Optical Interconnects XXIV*, pp. 17–20, Mar. 2024.

[C6] Chenghao Feng, **Shupeng Ning**, Jiaqi Gu, Hanqing Zhu, David Z Pan, and Ray T Chen. "Integrated Photonics for Computing and Artificial Intelligence". *2023 IEEE Photonics Society Summer Topicals Meeting Series (SUM)*, pp. 1–2, Jul. 2023.

[C5] **Shupeng Ning**, Hao-Chen Chang, Kang-Chieh Fan, Po-Yu Hsiao, and Ray T Chen. "A Lab-on-a-Chip Sensing Platform Enabling Concurrent Detection Using Subwavelength Grating Micro-Ring Resonator". *CLEO: Applications and Technology*, AW3Q–6, May 2023.

[C4] **Shupeng Ning**, Hao-Chen Chang, Kang-Chieh Fan, May H Hlaing, Sourabh Jain, Jackson Carmichael, Lauren Head, Devangshu Goswami, Sasid Sriwattana, Hayden Pietsch, et al. "Silicon Photonic Chip-Based Biosensor for COVID-19 and Flu Detection with High Sensitivity and Specificity". *Ultra-high-definition imaging systems VI*, pp. 26–28, Mar. 2023.

[C3] **Shupeng Ning**, Chao Wang, Hao-Chen Chang, Kyoung Min Yoo, James Fan, Devan Shoemaker, Maxwell Nakos, May H Hlaing, Yen-Wen Lu, Huiping Tian, et al. "A Point-of-Care Biosensor with Subwavelength Grating Waveguide-Based Micro-Ring Resonator for Detection of COVID-19". *CLEO: Science and Innovations*, JW3B–195, May 2022.

[C2] Xingli Xu, Xingchen Li, **Shupeng Ning**, Weiwei Cui, and Mark A Reed. "GHz Bulk Acoustic-Wave Resonator Array Actuated Minimized Collector for High-Efficiency E. coli Enrichment". *2021 IEEE International Ultrasonics Symposium (IUS)*, pp. 1–4, Sep. 2021.

[C1] Guanyu Zhang, Shuchang Liu, **Shupeng Ning**, Weiwei Cui, and Mark Reed. "Parallel Trapping and Controllable Lysis of Cells Using Acoustic Pillar Array Chip". *2021 IEEE International Ultrasonics Symposium (IUS)*, pp. 1–3, Sep. 2021.

PATENTS

[P1] Guohua Liu, **Shupeng Ning**, Wenjia Zhang, Qian Du, and Quan Zhang. "A Microfluidic Chip for Parallel Detection in Dark Field". Chinese Patent CN114471753B. Aug. 2023.

PROFESSIONAL SERVICE

Journal Reviewer

- **Nature Portfolio**: Nature Communications, Scientific Reports
- **Optica (formerly OSA)**: Optica, Journal of Lightwave Technology, Optics Express, Journal of Optical Communications and Networking
- **Elsevier**: Biosensors and Bioelectronics, Optics and Laser Technology, Optik
- **MDPI**: Biosensors, Micromachines, Chemosensors